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Faculty of Computer Science and Engineering

CC02 — Lab Report

Digital Signal Processing
Lab 5

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1 Lab 5

1.1 Overview

This is the report of lab 5.

1.2 Objectives

- Analyze systems using Z-transform, Z^--tranform, Z^+-Transform.

1.3 Methodologies

1.3.1 Hardware & Software

- Hardware: Intel-based Laptop
- Software: Scilab (Windows OS)

1.3.2 The Z-Transform

The z-transform of a discrete-time signal $x(n)$ is defined as the power series

$$X(z) = \sum_{n=-\infty}^{+\infty} x(n)z^{-n}$$

where z is a complex variable ($z = a + jb$ or $z = re^{j\delta}$). For convenience, the z-transform of a signal $x(n)$ is denoted by $X(z) = \mathcal{Z}\{x(n)\}$. The relationship between $x(n)$ and $X(z)$ is indicated by $x(n) \overset{z}{\longleftrightarrow} X(z)$. Furthermore, the region of convergence (ROC) of $X(z)$ is defined as the set of all values of z for which $X(z)$ attains a finite value, mathematically expressed as ROC: $\{z \mid |X(z)| < \infty\}$.

1.3.2.a Properties of the z-Transform

The z-transform exhibits several fundamental properties. The first is linearity: if $x_1(n) \longleftrightarrow X_1(z)$ with ROC: $X_1(z)$ and $x_2(n) \overset{\mathcal{Z}}{\longleftrightarrow} X_2(z)$ with ROC: $X_2(z)$, then a linear combination behaves as follows:

$$x(n) = ax_1(n) + bx_2(n) \longleftrightarrow X(z) = aX_1(z) + bX_2(z)$$

The resulting ROC is the intersection of the individual ROCs, such that ROC: $X(z) =$ ROC: $X_1(z) \cap$ ROC: $X_2(z)$.

Regarding time shifting, if $x(n) \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$ with ROC: $X(z)$, then the shifted signal transforms as $x(n-k) \overset{\mathcal{Z}}{\longleftrightarrow} z^{-k}X(z)$.

For scaling in the z-domain, if $x(n) \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$ with an ROC of $r_1 < |z| < r_2$, then scaling the sequence by a^n yields $a^n x(n) \overset{\mathcal{Z}}{\longleftrightarrow} X(a^{-1}z)$. This results in an ROC of $|a|r_1 < |z| < |a|r_2$, where a can be a real or complex value.

Time reversal dictates that if $x(n) \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$ with an ROC of $r_1 < |z| < r_2$, then $x(-n) \overset{\mathcal{Z}}{\longleftrightarrow} X(z^{-1})$, modifying the ROC to $\frac{1}{r_2} < |z| < \frac{1}{r_1}$.

Differentiation in the z-domain states that if $x(n) \overset{\mathcal{Z}}{\longleftrightarrow} X(z)$ with an ROC of $r_1 < |z| < r_2$, then multiplying the sequence by n transforms as $n \cdot x(n) \longleftrightarrow -z\frac{dX(z)}{dz}$, maintaining the same ROC of $r_1 < |z| < r_2$.

Finally, for the convolution of two sequences, if $x_1(n) \longleftrightarrow X_1(z)$ with ROC: $X_1(z)$ and $x_2(n) \longleftrightarrow X_2(z)$ with ROC: $X_2(z)$, then their convolution transforms as a product in the z-domain:

$$x(n) = x_1(n) * x_2(n) \longleftrightarrow X(z) = X_1(z)X_2(z)$$

with the ROC being the intersection of the individual ROCs, ROC: $X(z)$ = ROC: $X_1(z) \cap$ ROC: $X_2(z)$.

1.3.2.b Rational z-Transform

For a rational z-transform, the zeros of $X(z)$ are defined as the values of z for which $X(z) = 0$. Conversely, the poles of $X(z)$ are the values of z for which $X(z) = \infty$.

1.4 Exercise

For each discrete-time signal $x(n)$, find the corresponding Z-Transform and its ROC.

1.4.1 Exercise 1

Use Z-transform to find the impulse response $h(n)$ of the system represented by the following input-output description equation:

$$y(n) - y(n-2) = x(n)$$

Solution

The impulse response $h(n)$ is found using the Z-transform. Starting with the difference equation and taking the Z-transform of both sides:

$$Y(z) - z^{-2}Y(z) = X(z)$$

$$Y(z)(1 - z^{-2}) = X(z)$$

$$\Rightarrow H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - z^{-2}} = \frac{1}{1^2 - (z^{-1})^2}$$

Partial fraction decomposition gives:

$$H(z) = \frac{1}{2}\left(\frac{1}{1 - z^{-1}} + \frac{1}{1 + z^{-1}}\right)$$

Taking the inverse Z-transform:

$$h(n) = \frac{1}{2}(1^n + (-1)^n)u(n)$$

$$h(n) = \frac{1 + (-1)^n}{2}u(n)$$

ROC: $|z| > 1$

1.4.2 Exercise 2

Use Z and Z^{-1} transform to compute the convolution for the following pairs:

1. $x_1(n) = \{\underset{\uparrow}{1}, 2, 3, 4, 5\}$ and $x_2(n) = \{\underset{\uparrow}{1}, 1, 1\}$
2. $x_1(n) = \left(\frac{1}{5}\right)^n u(n)$ and $x_2(n) = 2^n u(n)$
3. $x_1(n) = nu(n)$ and $x_2(n) = 2^n u(n-1)$

Solution

a. Convolution of $x_1(n) = \{\underset{\uparrow}{1}, 2, 3, 4, 5\}$ and $x_2(n) = \{\underset{\uparrow}{1}, 1, 1\}$

Display the signal in term of $\delta(n)$:

$$x_1(n) = 1\delta(n) + 2\delta(n-1) + 3\delta(n-2) + 4\delta(n-3) + 5\delta(n-4)$$

$$x_2(n) = 1\delta(n) + 1\delta(n-1) + 1\delta(n-2)$$

Using Z-transforms:

$$X_1(z) = 1 + 2z^{-1} + 3z^{-2} + 4z^{-3} + 5z^{-4} \quad \text{All } z \text{ except 0 or } \infty$$

$$X_2(z) = 1 + z^{-1} + z^{-2} \quad \text{All } z \text{ except 0 or } \infty$$

Multiplying and combining terms:

$$Y(z) = X_1(z) \cdot X_2(z) = 1 + 3z^{-1} + 6z^{-2} + 9z^{-3} + 12z^{-4} + 9z^{-5} + 5z^{-6}, \quad \text{All } z \text{ except 0 or } \infty$$

$$y(n) = \{1 \uparrow, 3, 6, 9, 12, 9, 5\}$$

—

b. Convolution of $x_1(n) = \left(\frac{1}{5}\right)^n \cdot u(n)$ and $x_2(n) = 2^n \cdot u(n)$

Display the signal in Z-transforms:

$$X_1(z) = \frac{1}{1 - \frac{1}{5}z^{-1}} \quad \text{with } |z| > \frac{1}{5}$$

$$X_2(z) = \frac{1}{1 - 2z^{-1}} \quad \text{with } |z| > 2$$

Multiplying in Z domains:

$$Y(z) = X_1(z) \cdot X_2(z) = \frac{1}{1 - \frac{1}{5}z^{-1}} \cdot \frac{1}{1 - 2z^{-1}}, \quad |z| > 2$$

Partial fractions of $Y(z)$:

$$Y(z) = A \cdot \frac{1}{1 - 2z^{-1}} - B \cdot \frac{1}{1 - \frac{1}{5}z^{-1}} = \frac{1}{\left(1 - \frac{1}{5}z^{-1}\right) \cdot (1 - 2z^{-1})} \quad \text{with } |z| > 2$$

$$Y(z) = \frac{10}{9} \cdot \frac{1}{1 - 2z^{-1}} - \frac{1}{9} \cdot \frac{1}{1 - \frac{1}{5}z^{-1}} \quad \text{with } |z| > 2$$

Inverse Z-transform:

$$y(n) = \left(\frac{10}{9} \cdot 2^n\right) \cdot u(n) - \left(\frac{1}{9}\left(\frac{1}{5}\right)^n\right) \cdot u(n)$$

c. Convolution of $x_1(n) = nu(n)$ and $x_2(n) = 2^n u(n-1)$

Using the standard Z-transform pair for a ramp sequence:

Given $x_1(n) = nu(n)$.

$$X_1(z) = \frac{z^{-1}}{(1 - z^{-1})^2} = \frac{z}{(z-1)^2} \quad \text{with ROC: } |z| > 1$$

Given $x_2(n) = 2^n u(n-1)$.

We can rewrite this to match standard transform pairs. Let's factor out a 2:

$$x_2(n) = 2 \cdot 2^{n-1} u(n-1)$$

We know the standard transform for $a^n u(n) \overset{\mathcal{Z}}{\leftrightarrow} \frac{z}{z-a}$. Using the time-delay property $x(n-k) \overset{\mathcal{Z}}{\leftrightarrow} z^{-k} X(z)$, where $k = 1$ and $a = 2$:

$$2^{n-1} u(n-1) \overset{\mathcal{Z}}{\leftrightarrow} z^{-1} \left(\frac{z}{z-2} \right) = \frac{1}{z-2}$$

Multiplying by the constant 2 gives us $X_2(z)$:

$$X_2(z) = \frac{2}{z-2}$$

ROC: $|z| > 2$

Multiplying in Z domains:

$$Y(z) = \left(\frac{z}{(z-1)^2} \right) \cdot \left(\frac{2}{z-2} \right) = \frac{2z}{(z-1)^2(z-2)}$$

ROC: The intersection of the two ROCs is $|z| > 2$.

Partial fractions of $Y(z)$:

$$\frac{Y(z)}{(z)} = \frac{2z}{(z-1)^2(z-2)} = \frac{A}{z-1} + \frac{B}{(z-1)^2} + \frac{C}{z-2}$$

Finding coefficients:

With $z = 1$, multiplies with $(z-1)^2$:

$$B = \frac{2}{1-2} = -2$$

With $z = 1$, multiplies with $(z-1)$:

$$C = \frac{2}{(2-1)^2} = 2$$

With $B = -2$, $C = 2$, evaluate at $z = 0$:

$$\frac{2}{(-1)^2(-2)} = \frac{A}{-1} + \frac{-2}{(-1)^2} + \frac{2}{-2}$$

$$\implies A = -2$$

Substitute the constants back into the expansion:

$$\frac{Y(z)}{z} = \frac{-2}{z-1} - \frac{2}{(z-1)^2} + \frac{2}{z-2}$$

$$Y(z) = -2\left(\frac{z}{z-1}\right) - 2\left(\frac{z}{(z-1)^2}\right) + 2\left(\frac{z}{z-2}\right)$$

Applying inverse Z-transform:

$$\mathcal{Z}^{-1}\left\{\frac{z}{z-1}\right\} = u(n)$$

$$\mathcal{Z}^{-1}\left\{\frac{z}{(z-1)^2}\right\} = nu(n)$$

$$\mathcal{Z}^{-1}\left\{\frac{z}{z-2}\right\} = 2^n u(n)$$

$$\implies y(n) = -2u(n) - 2nu(n) + 2(2^n)u(n)$$

$$y(n) = \left[2^{n+1} - 2n - 2\right] u(n)$$

1.4.3 Exercise 3

Find all possible $x(n)$ that have the following Z^{-1} transforms:

1. $X_1(z) = \frac{1}{2-3z^{-1}+z^{-2}}$
2. $X_2(z) = \frac{1+2z^{-1}+z^{-2}}{1+4z^{-1}+4z^{-2}}$
3. $X_3(z) = \frac{1+z^{-1}}{(1-0.5z^{-1})^2(1-0.3z^{-1})}, \quad 0.3 < |z| < 0.5$

Solution

Find all possible $x(n)$ that has Z^{-1} transform as follows:

Problem 3.1:

$$X_1(z) = \frac{1}{2-3z^{-1}+z^{-2}}$$

$$X_1(z) = \frac{1}{2-2z^{-1}-z^{-1}+z^{-2}}$$

$$X_1(z) = \frac{1}{(1-z^{-1})\cdot(2-z^{-1})}$$

Partial Fraction Decomposition:

$$\frac{X_1(z)}{z} = \frac{z^2}{z \cdot (z-1) \cdot (2z-1)}$$

$$\frac{X_1(z)}{z} = \frac{z}{(z-1) \cdot (2z-1)}$$

Decompose process:

$$\frac{X_1(z)}{z} = \frac{A}{(z-1)} + \frac{B}{(2z-1)}$$

We have:

$$A \cdot (2z-1) + B(z-1) = z$$

Put $z = 1$:

$$A \cdot (2 \cdot 1 - 1) + B \cdot (1-1) = 1 \implies A = 1$$

Put $z = \frac{1}{2}$:

$$A \cdot \left(2 \cdot \frac{1}{2} - 1\right) + B \cdot \left(\frac{1}{2} - 1\right) = \frac{1}{2} \implies B = -1$$

So, in conclusion we have:

$$\frac{X_1(z)}{z} = \frac{1}{(z-1)} + \frac{-1}{(2z-1)}$$

$$X_1(z) = \frac{z}{z-1} - \frac{1/2 \cdot z}{z - \frac{1}{2}}$$

Divide numerator and denominator by z:

$$X_1(z) = \frac{1}{1 - z^{-1}} - \frac{1/2}{1 - \frac{1}{2}z^{-1}}$$

With $X_1(z)$, we have 3 cases based on its ROC:

Case 1: Causal $|z| > 1$

$$x(n) = u(n) - \frac{1}{2} \cdot \left(\frac{1}{2}\right)^n \cdot u(n)$$

Case 2: Mixed $\frac{1}{2} < |z| < 1$

$$x(n) = -u(-n-1) - \frac{1}{2} \cdot \left(\frac{1}{2}\right)^n \cdot u(n)$$

Case 3: Anti-causal $|z| < \frac{1}{2}$

$$x(n) = -u(-n-1) - \frac{1}{2} \cdot \left(-\frac{1}{2}\right)^n \cdot u(-n-1)$$

Problem 3.2

$$X_2(z) = \frac{1 + 2z^{-1} + z^{-2}}{1 + 4z^{-1} + 4z^{-2}}$$

$$X_2(z) = \frac{z^2 + 2z + 1}{z^2 + 4z + 4}$$

$$X_2(z) = \frac{z^2 + 2z + 1}{(z+2)^2}$$

Partial Fraction Decomposition:

$$\frac{X_2(z)}{z} = \frac{z^2 + 2z + 1}{z \cdot (z+2)^2}$$

Decompose process:

$$\frac{X_2(z)}{z} = \frac{A}{z} + \frac{B}{(z+2)} + \frac{C}{(z+2)^2}$$

We have:

$$A \cdot (z+2)^2 + B \cdot z \cdot (z+2) + C \cdot z = z^2 + 2z + 1$$

Put $z = -2$:

$$-2C = 1 \implies C = -\frac{1}{2}$$

Put $z = 0$:

$$4A = 1 \implies A = \frac{1}{4}$$

Put $z = 1$:

$$9A + 3B + C = 4 \implies B = \frac{3}{4}$$

So, in conclusion we have:

$$\frac{X_2(z)}{z} = \frac{1/4}{z} + \frac{3/4}{(z+2)} - \frac{1/2}{(z+2)^2}$$

$$X_2(z) = \frac{1/4 \cdot z}{z} + \frac{3/4 \cdot z}{(z+2)} - \frac{1/2 \cdot z}{(z+2)^2}$$

Divide numerator and denominator by z:

$$X_2(z) = \frac{1}{4} + \frac{3/4}{(1+2z^{-1})} - \frac{1/2 \cdot z^{-1}}{(1+2z^{-1})^2}$$

With $X_2(z)$, we have 2 cases based on its ROC:

Case 1: Causal $|z| > |2|$

$$x(n) = \frac{1}{4} \cdot \delta(n) + \frac{3}{4} \cdot (-2)^n \cdot u(n) + \frac{1}{4} \cdot n \cdot (-2)^n \cdot u(n)$$

Case 2: Non-causal $|z| < |2|$

$$x(n) = \frac{1}{4} \cdot \delta(n) + \frac{3}{4} \cdot 2^n \cdot u(-n-1) + \frac{1}{4} \cdot n \cdot 2^n \cdot u(-n-1)$$

Problem 3.3

$$X_3(z) = \frac{1+z^{-1}}{(1-0.5z^{-1})^2 \cdot (1-0.3z^{-1})}$$

$$X_3(z) = \frac{(1+z^{-1}) \cdot z^3}{(1-0.5z^{-1})^2 \cdot z^2 \cdot (1-0.3z^{-1}) \cdot z}$$

$$X_3(z) = \frac{z^3 + z^2}{(z-0.5)^2 \cdot (z-0.3)}$$

Partial Fraction Decomposition:

$$\frac{X_3(z)}{z} = \frac{z^2(z+1)}{z \cdot (z-0.5)^2 \cdot (z-0.3)}$$

$$\frac{X_3(z)}{z} = \frac{z(z+1)}{(z-0.5)^2 \cdot (z-0.3)}$$

Decompose process:

$$\frac{X_3(z)}{z} = \frac{A}{(z-0.5)^2} + \frac{B}{(z-0.5)} + \frac{C}{(z-0.3)}$$

We have:

$$A \cdot (z-0.3) + B \cdot (z-0.3) \cdot (z-0.5) + C \cdot (z-0.5)^2 = z(z+1)$$

Put $z = 0.5$:

$$0.2A = \frac{3}{4} \implies A = \frac{15}{4}$$

Put $z = 0.3$:

$$\frac{C}{25} = \frac{39}{100} \implies C = \frac{39}{4}$$

Put $z = 0$:

$$-\frac{3}{10}A + \frac{3}{20}B + \frac{1}{4}C = 0 \implies B = \frac{-35}{4}$$

So, in conclusion we have:

$$\frac{X_3(z)}{z} = \frac{15/4}{(z-0.5)^2} - \frac{35/4}{(z-0.5)} + \frac{39/4}{(z-0.3)}$$

$$X_3(z) = \frac{15/4 \cdot z}{(z-0.5)^2} - \frac{35/4 \cdot z}{(z-0.5)} + \frac{39/4 \cdot z}{(z-0.3)}$$

Divide numerator and denominator by z or z^2:

$$X_3(z) = \frac{15/4 \cdot z^{-1}}{(1-0.5z^{-1})^2} - \frac{35/4}{(1-0.5z^{-1})} + \frac{39/4}{(1-0.3z^{-1})}$$

Based on the ROC of topic $0.3 < |z| < 0.5$, we have:

$$x_3(n) = -\frac{15}{2} \cdot n \cdot \left(\frac{1}{2}\right)^n \cdot u(-n-1) + \frac{35}{4} \cdot \left(\frac{1}{2}\right)^n \cdot u(-n-1) + \frac{39}{4} \cdot \left(\frac{3}{10}\right)^n \cdot u(n)$$

1.4.4 Exercise 4

Given an LTI system defined by the following input-output description equation:

$$y(n) = 0.7y(n-1) + x(n)$$

1. Draw the block diagram of the above system.
2. Determine $h(n)$.
3. Determine $y(n)$ when $x(n) = u(n)$.

Solution

a. Draw the block diagram of the above system

b. Determine $h(n)$

To determine $h(n)$, we will first transform $y(n)$ and $x(n)$ to Z form. Then we will find $H(z)$ by equation:

$$H(z) = \frac{Y(z)}{X(z)}$$

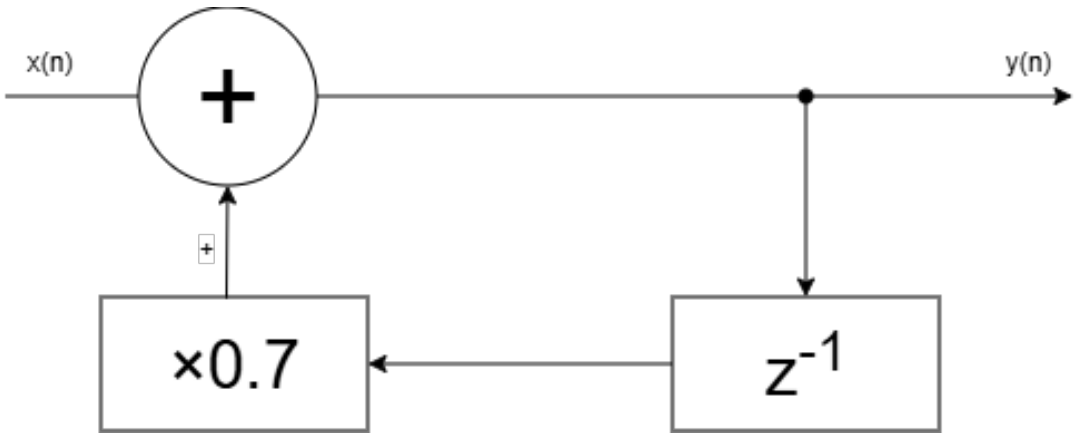

Figure 1: *Block Diagram*

After that, we will find $h(n)$ by using inverse Z-transform of $H(z)$.

Transform to Z form:

$$Y(z) = 0.7 \cdot z^{-1} \cdot Y(z) + X(z) \iff Y(z) = \frac{X(z)}{1 - 0.7z^{-1}}$$

$$x(z) = X(z)$$

So:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\frac{X(z)}{1-0.7z^{-1}}}{X(z)} = \frac{1}{1 - 0.7z^{-1}}$$

Inverse Z transform: For ROC: $|z| > 0.7$

$$h(n) = 0.7^n u(n)$$

For ROC: $|z| < 0.7$

$$h(n) = -0.7^n u(-n-1)$$

c. Determine $y(n)$ when $x(n) = u(n)$ When $x(n) = u(n)$: With $H(z)$ which we have defined in b, we will find new $Y(z)$ by inverse Z transform from $Y(z)$ which will be determined by equation:

$$Y(z) = X(z) \cdot H(z)$$

Transform $x(n)$ to Z form:

$$X(z) = \frac{1}{1 - z^{-1}}$$

Multiply:

$$Y(z) = H(z) \cdot X(z) = \frac{1}{(1 - 0.7z^{-1}) \cdot (1 - z^{-1})}$$

Partial Fraction Decomposition:

$$Y(z) = \frac{z^2}{(z-0.7)\cdot(z-1)}$$

$$\frac{Y(z)}{z} = \frac{z}{(z-0.7)\cdot(z-1)}$$

Decompose process:

$$\frac{Y(z)}{z} = \frac{A}{(z-0.7)} + \frac{B}{(z-1)}$$

We have:

$$A(z-1) + B(z-0.7) = z$$

Put $z = 1$:

$$0.3B = 1 \implies B = \frac{10}{3}$$

Put $z = 0.7$:

$$-0.3A = 0.7 \implies A = -\frac{7}{3}$$

So, in conclusion we have:

$$\frac{Y(z)}{z} = \frac{10/3}{z-0.7} - \frac{7/3}{z-1}$$

$$Y(z) = \frac{-7/3\cdot z}{z-0.7} + \frac{10/3\cdot z}{z-1}$$

$$Y(z) = \frac{-7/3}{1-0.7z^{-1}} + \frac{10/3}{1-z^{-1}}$$

Inverse Z transform to find $y(n)$: For ROC: $|z| > 1$

$$y(n) = -\frac{7}{3}\cdot(0.7)^n\cdot u(n) + \frac{10}{3}\cdot u(n)$$

For ROC: $0.7 < |z| < 1$

$$y(n) = -\frac{7}{3}\cdot(0.7)^n\cdot u(n) - \frac{10}{3}\cdot u(-n-1)$$

For ROC: $|z| < 0.7$

$$y(n) = \frac{7}{3}\cdot(0.7)^n\cdot u(-n-1) - \frac{10}{3}\cdot u(-n-1)$$

1.4.5 Exercise 5

Given an LTI system defined by the following input-output description equation:

$$y(n) = 2y(n-1) - 3y(n-2) + x(n) + x(n-1)$$

1. Draw the block diagram of the above system.
2. Determine the impulse response $h(n)$.
3. Determine $y_{zi}(n)$ when $y(-1) = y(-2) = 1$.
4. Determine $y_{zs}(n)$ when $x(n) = 2^n u(n)$.

Solution

a. Draw the block diagram of the above system

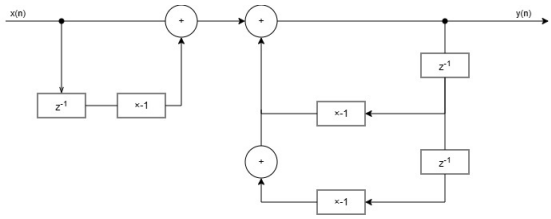

Figure 2: *Block Diagram*

b. Determine the impulse response $h(n)$

To determine $h(n)$, we will first transform $y(n)$ and $x(n)$ to Z form. Then we will find $H(z)$ by equation:

$$H(z) = \frac{Y(z)}{X(z)}$$

After that, we will find $h(n)$ by using inverse Z-transform of $H(z)$.

Transform to Z form:

$$Y(z) = 2z^{-1} \cdot Y(z) - 3z^{-2} \cdot Y(z) + X(z) + z^{-1} \cdot X(z) \iff Y(z) = \frac{X(z)(1+z^{-1})}{1-2z^{-1}+3z^{-2}}$$

$$x(z) = X(z)$$

So:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\frac{X(z)(1+z^{-1})}{1-2z^{-1}+3z^{-2}}}{X(z)} = \frac{1+z^{-1}}{1-2z^{-1}+3z^{-2}}$$

Partial Fraction Decomposition:

$$H(z) = \frac{z^2+z}{z^2-2z+3}$$

$$\frac{H(z)}{z} = \frac{z+1}{z^2-2z+3}$$

Solve the equation to get poles of z: This system has two complex-conjugate poles at z, $H(z)$ will become:

$$\frac{H(z)}{z} = \frac{A}{z-p_1} + \frac{B}{z-p_2}$$

With:

$$p_1 = (1 + j\sqrt{2})$$

$$p_2 = (1 - j\sqrt{2})$$

To obtain A and B, we follow formula:

$$A = \frac{(z-p_1)H(z)}{z}\bigg|_{z=p_1} = \frac{z+1}{z-p_2}\bigg|_{z=p_1} = \frac{1+j\sqrt{2}+1}{1+j\sqrt{2}-1+j\sqrt{2}} = \frac{1}{2} - j\frac{\sqrt{2}}{2}$$

$$B = \frac{(z-p_2)H(z)}{z}\bigg|_{z=p_2} = \frac{z+1}{z-p_1}\bigg|_{z=p_2} = \frac{1-j\sqrt{2}+1}{1-j\sqrt{2}-1-j\sqrt{2}} = \frac{1}{2} + j\frac{\sqrt{2}}{2}$$

Conclusion:

$$\frac{H(z)}{z} = \frac{\frac{1}{2} - j\frac{\sqrt{2}}{2}}{z-(1+j\sqrt{2})} + \frac{\frac{1}{2} + j\frac{\sqrt{2}}{2}}{z-(1-j\sqrt{2})}$$

$$H(z) = \frac{\frac{1}{2} - j\frac{\sqrt{2}}{2}}{1-(1+j\sqrt{2})z^{-1}} + \frac{\frac{1}{2} + j\frac{\sqrt{2}}{2}}{1-(1-j\sqrt{2})z^{-1}}$$

Inverse Z-transform by following the formula in theory:

$$h(n) = \sqrt{3} \cdot (\sqrt{3})^n \cdot \cos(\arctan(\sqrt{2})n + 54.73^\circ) \cdot u(n)$$

c. Determine $y_{zi}(n)$ when $y(-1) = y(-2) = 1$

The two poles of z is:

$$p_1 = \sqrt{3} \cdot e^{j \cdot \theta}$$

$$p_2 = \sqrt{3} \cdot e^{-j \cdot \theta}$$

In the a, we have $H(z)$ is:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{B(z)}{A(z)} = \frac{1+z^{-1}}{1-2z^{-1}+3z^{-2}}$$

For the initial conditions $y(-1) = y(-2) = 1$ the additional component in the z-transform is:

$$Y_{zi}(z) = \frac{N_o(z)}{A(z)} = \frac{-1-3z^{-1}}{1-2z^{-1}+3z^{-2}}$$

Decompose process:

$$Y_{zi}(z) = \frac{-z^2-3z}{z^2-2z+3z^2}$$

$$\frac{Y_{zi}(z)}{z} = \frac{-z-3}{z^2-2z+3}$$

Now, we must to make inverse Z-transform to find $y_{zi}(n)$. Solve the equation to get poles of z:

This system has two complex-conjugate poles at z, $Y_{zi}(z)$ will become:

$$\frac{Y_{zi}(z)}{z} = \frac{A}{z-p_1} + \frac{B}{z-p_2}$$

With:

$$p_1 = (1 + j\sqrt{2})$$

$$p_2 = (1 - j\sqrt{2})$$

To obtain A and B, we follow formula:

$$A = \frac{(z-p_1)\cdot Y_{zi}(z)}{z}\bigg|_{z=p_1} = \frac{-z-3}{z-p_2}\bigg|_{z=p_1} = \frac{-1-j\sqrt{2}-3}{1+j\sqrt{2}-1+j\sqrt{2}} = -\frac{1}{2} + j\sqrt{2}$$

$$B = \frac{(z-p_2)\cdot Y_{zi}(z)}{z}\bigg|_{z=p_2} = \frac{-z-3}{z-p_1}\bigg|_{z=p_2} = \frac{-1+j\sqrt{2}-3}{1-j\sqrt{2}-1-j\sqrt{2}} = -\frac{1}{2} - j\sqrt{2}$$

Conclusion:

$$\frac{Y_{zi}(z)}{z} = \frac{-\frac{1}{2}+j\sqrt{2}}{z-(1+j\sqrt{2})} + \frac{-\frac{1}{2}-j\sqrt{2}}{z-(1-j\sqrt{2})}$$

$$Y_{zi}(z) = \frac{-\frac{1}{2}+j\sqrt{2}}{1-(1+j\sqrt{2})z^{-1}} + \frac{-\frac{1}{2}-j\sqrt{2}}{1-(1-j\sqrt{2})z^{-1}}$$

Inverse Z-transform by following the formula in theory:

$$y_{zi}(n) = 3\cdot(\sqrt{3})^n \cdot \cos(\arctan(\sqrt{2})n - 70.53^\circ)\cdot u(n)$$

d. Determine $y_{zs}(n)$ when $x(n) = 2^n u(n)$

The Z-transform of $x(n)$: ROC: $|z| > 2$

$$X(z) = \frac{1}{1-2z^{-1}}$$

Followed the formula of $Y_{zs}(z) = H(z)\cdot X(z)$, we have:

$$Y_{zs}(z) = H(z)\cdot X(z)$$

$$Y_{zs}(z) = \frac{1+z^{-1}}{(1-2z^{-1}+3z^{-2})\cdot(1-2z^{-1})}$$

$$Y_{zs}(z) = \frac{z^3+z^2}{(z^2-2z+3)\cdot(z-2)}$$

$$\frac{Y_{zs}(z)}{z} = \frac{z^2+z}{(z^2-2z+3)\cdot(z-2)}$$

We will separate it into 2 parts:

$$\frac{Y_{zs}(z)}{z} = \frac{3-z}{z^2-2z+3} + \frac{2}{z-2}$$

Decompose process of part 1: This system has two complex-conjugate poles at z:

$$\frac{3-z}{z^2-2z+3} = \frac{A}{z-p_1} + \frac{B}{z-p_2}$$

With:

$$p_1 = (1+j\sqrt{2})$$

$$p_2 = (1-j\sqrt{2})$$

To obtain A and B, we follow formula:

$$A = \left.\frac{(z-p_1)\cdot Y_{zs}(z)}{z}\right|_{z=p_1} = \left.\frac{3-z}{z-p_2}\right|_{z=p_1} = \frac{3-1-j\sqrt{2}}{1+j\sqrt{2}-1+j\sqrt{2}} = -\frac{1}{2} - j\frac{\sqrt{2}}{2}$$

$$B = \left.\frac{(z-p_2)\cdot Y_{zs}(z)}{z}\right|_{z=p_2} = \left.\frac{3-z}{z-p_1}\right|_{z=p_2} = \frac{3-1+j\sqrt{2}}{1-j\sqrt{2}-1-j\sqrt{2}} = -\frac{1}{2} + j\frac{\sqrt{2}}{2}$$

Conclusion:

$$\frac{Y_{zs}(z)}{z} = \frac{-\frac{1}{2} - j\frac{\sqrt{2}}{2}}{z-(1+j\sqrt{2})} + \frac{-\frac{1}{2} + j\frac{\sqrt{2}}{2}}{z-(1-j\sqrt{2})} + \frac{2}{z-2}$$

$$Y_{zs}(z) = \frac{-\frac{1}{2} - j\frac{\sqrt{2}}{2}}{1-(1+j\sqrt{2})z^{-1}} + \frac{-\frac{1}{2} + j\frac{\sqrt{2}}{2}}{1-(1-j\sqrt{2})z^{-1}} + \frac{2}{1-2z^{-1}}$$

Inverse Z-transform by following the formula in theory:

$$y_{zs}(n) = \sqrt{3}\cdot(\sqrt{3})^n \cdot \cos(\arctan(\sqrt{2})n + 125.26^\circ)\cdot u(n) + 2\cdot 2^n \cdot u(n)$$

1.4.6 Exercise 6

If $x(n) \xrightarrow{\mathcal{Z}} X(z)$, prove the following statements:

1. $\mathcal{Z}\{x^*(n)\} = X^*(z^*)$
2. $\mathcal{Z}\{\Re e[x(n)]\} = \frac{1}{2}[X(z) + X^*(z^*)]$
3. $\mathcal{Z}\{\Im m[x(n)]\} = \frac{1}{2j}[X(z) - X^*(z^*)]$
4. $\mathcal{Z}\{e^{j\omega_0 n}x(n)\} = X(ze^{-j\omega_0})$

Solution

a. $\mathcal{Z}[x^*(n)] = X^*(z^*)$

$$\mathcal{Z}[x^*(n)] = \sum_{n=-\infty}^{\infty} x^*(n) \cdot z^{-n} = \sum_{n=-\infty}^{\infty} [x(n) \cdot (z^*)^n]^* = X^*(z^*)$$

b. $\mathcal{Z}\{\mathbf{Re}[x(n)]\} = \frac{1}{2}[X(z) + X^*(z^*)]$

Base on the solution in previous part, we have:

$$\frac{1}{2}[X(z) + X^*(z^*)] = \mathcal{Z}\left[\frac{x(n) + x^*(n)}{2}\right]$$

$$= \mathcal{Z}\left[\frac{a + bj + a - bj}{2}\right] = \mathcal{Z}[a] = \mathcal{Z}[\mathrm{Re}[x(n)]]$$

c. $\mathcal{Z}\{\mathbf{Im}[x(n)]\} = \frac{1}{2j}[X(z) - X^*(z^*)]$

Base on the solution in previous part, we have:

$$\frac{1}{2j}[X(z) - X^*(z^*)] = \mathcal{Z}\left[\frac{x(n) - x^*(n)}{2j}\right]$$

$$= \mathcal{Z}\left[\frac{a + bj - a + bj}{2j}\right] = \mathcal{Z}[b] = \mathcal{Z}[\mathrm{Im}[x(n)]]$$

d. $\mathcal{Z}\{e^{j\omega_0 n}x(n)\} = X(ze^{-j\omega_0})$

$$\sum_{n=-\infty}^{\infty} e^{j\omega_0 n}x(n)z^{-n} = \sum_{n=-\infty}^{\infty} x(n)(e^{-j\omega_0}z)^{-n}$$

$$= X(ze^{-j\omega_0})$$

1.5 Additional Exercises

Complete exercises 3.3, 3.4, 3.7, 3.8, 3.13, 3.14, 3.19 and 3.20 in the reference document in the attachment.

1.5.1 Exercise 3.3

Determine the z-transforms and sketch the ROC of the following signals.

(a) $x_1(n) = \begin{cases} (\frac{1}{3})^n, & n \geq 0 \\ (\frac{1}{2})^{-n}, & n < 0 \end{cases}$

(b) $x_2(n) = \begin{cases} (\frac{1}{3})^n - 2^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$

(c) $x_3(n) = x_1(n+4)$

(d) $x_4(n) = x_1(-n)$

Solution

(a) $x_1(n) = \begin{cases} (\frac{1}{3})^n, & n \geq 0 \\ (\frac{1}{2})^{-n}, & n < 0 \end{cases}$

Split the Z-Transform

$$X_1(z) = \underbrace{\sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^{-n}}_{X_1^+(z)} + \underbrace{\sum_{n=-\infty}^{-1} 2^n z^{-n}}_{X_1^-(z)}$$

Solve the Right-Sided Part ($n \geq 0$)

$$X_1^+(z) = \sum_{n=0}^{\infty} \left(\frac{1}{3} z^{-1}\right)^n = \frac{1}{1 - \frac{1}{3} z^{-1}}, \qquad |z| > \frac{1}{3}$$

Solve the Left-Sided Part ($n < 0$)

Let $m = -n$, so as $n : -1 \to -\infty$, we have $m : 1 \to \infty$:

$$X_1^-(z) = \sum_{m=1}^{\infty} 2^{-m} z^m = \sum_{m=1}^{\infty} \left(\frac{z}{2}\right)^m = \frac{z/2}{1 - z/2} = \frac{-1}{1 - 2z^{-1}}, \qquad |z| < 2$$

Combine

$$X_1(z) = \frac{1}{1 - \frac{1}{3} z^{-1}} - \frac{1}{1 - 2z^{-1}}, \qquad \frac{1}{3} < |z| < 2$$

Expressing as a single fraction (using $\frac{1}{1 - az^{-1}} = \frac{z}{z - a}$):

$$X_1(z) = \frac{z}{z - \frac{1}{3}} - \frac{z}{z - 2} = \frac{z(z-2) - z\left(z - \frac{1}{3}\right)}{\left(z - \frac{1}{3}\right)(z - 2)} = \frac{-\frac{5}{3} z}{\left(z - \frac{1}{3}\right)(z - 2)}$$

$$X_1(z) = \frac{-\frac{5}{3} z}{\left(z - \frac{1}{3}\right)(z - 2)}, \qquad \text{ROC: } \frac{1}{3} < |z| < 2$$

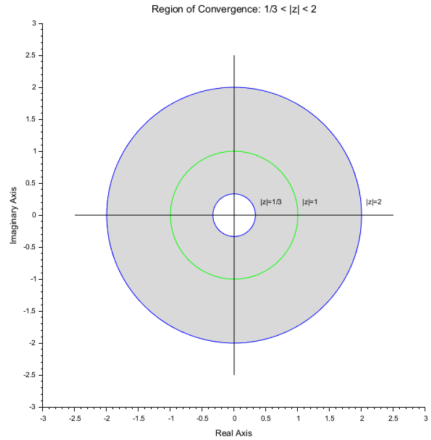

Figure 3: *ROC*

(b) $x_2(n) = \begin{cases} (\frac{1}{3})^n - 2^n, & n \geq 0 \\ 0, & n < 0 \end{cases}$

Since $x_2(n) = (\frac{1}{3})^n u[n] - 2^n u[n]$, its transform is:

$$X_2(z) = \sum_{n=0}^{\infty} [(\frac{1}{3})^n - 2^n] z^{-n} = \frac{1}{1 - \frac{1}{3}z^{-1}} - \frac{1}{1 - 2z^{-1}}$$

Each term requires $|z| > |a|$, so the overall ROC is:

$$\text{ROC: } |z| > 2$$

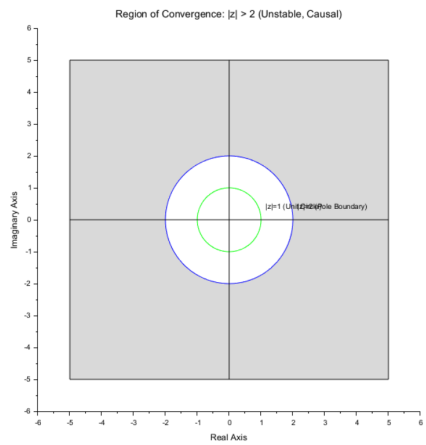

Figure 4: *ROC*

(c) $x_3(n) = x_1(n+4)$

A shift by $+4$ in n multiplies the Z-transform by z^4:

$$X_3(z) = z^4 X_1(z), \quad \text{ROC: } \frac{1}{3} < |z| < 2$$

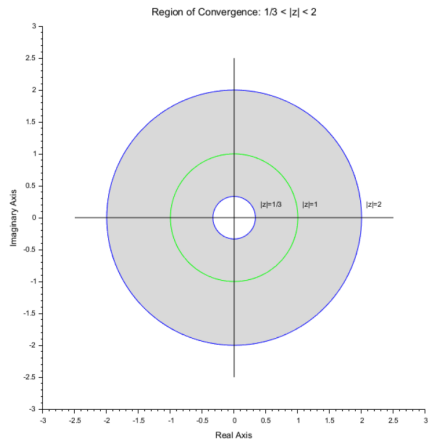

Figure 5: *ROC*

(d) $x_4(n) = x_1(-n)$

Time reversal $n \to -n$ replaces z by z^{-1} and inverts the ROC:

$$X_4(z) = X_1(z^{-1}) = \frac{1}{1 - \frac{1}{3}z} + \frac{z^{-1}}{2 - z^{-1}}$$

$$\text{ROC: } \frac{1}{2} < |z| < 3$$

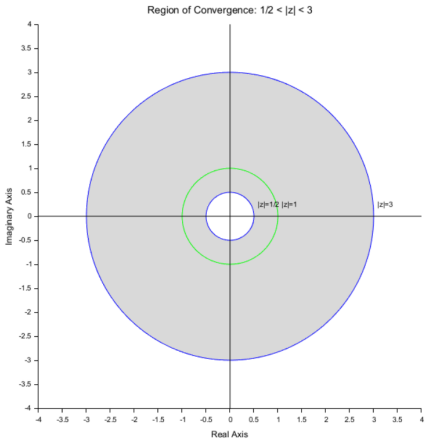

Figure 6: *ROC*

1.5.2 Exercise 3.4

Determine the z-transform of the following signals.

(a) $x(n) = n(-1)^n u(n)$

(b) $x(n) = n^2 u(n)$

(c) $x(n) = -na^n u(-n-1)$

(d) $x(n) = (-1)^n \left(\cos \frac{\pi}{3} n\right) u(n)$

(e) $x(n) = (-1)^n u(n)$

(f) $x(n) = \{\underset{\uparrow}{1}, 0, -1, 0, 1, -1, \dots\}$

Solution

(a) $x(n) = n(-1)^n u(n)$

$$X_1(z) = \sum_{n=0}^{\infty} n(-1)^n z^{-n} = \sum_{n=0}^{\infty} n(-z^{-1})^n$$

Let $r = -z^{-1}$. With the identity $\sum_{n=0}^{\infty} nr^n = \frac{r}{(1-r)^2}$ for $|r| < 1$:

$$X_1(z) = \frac{-z^{-1}}{(1+z^{-1})^2} = -\frac{z}{(z+1)^2}, \quad \text{ROC: } |z| > 1$$

(b) $x(n) = n^2 u(n)$

$$X_2(z) = \sum_{n=0}^{\infty} n^2 z^{-n}$$

Using the identity $\sum_{n=0}^{\infty} n^2 r^n = \frac{r(1+r)}{(1-r)^3}$ with $r = z^{-1}$:

$$X_2(z) = \frac{z^{-1}(1+z^{-1})}{(1-z^{-1})^3} = \frac{z^2+z}{(z-1)^3}, \quad \text{ROC: } |z| > 1$$

(c) $x(n) = -na^n u(-n-1)$

Here $x_3(n)$ is left-sided ($n \leq -1$). Put $k = -n$, so $k \geq 1$:

$$X_3(z) = \sum_{k=1}^{\infty} k \left(\frac{z}{a}\right)^k = \frac{\frac{z}{a}}{(1-\frac{z}{a})^2}, \quad \text{ROC: } |z| < |a|$$

(d) $x(n) = (-1)^n \left(\cos \frac{\pi}{3} n\right) u(n)$

Rewrite using Euler's formula:

$$(-1)^n \cos\left(\frac{\pi n}{3}\right) = e^{j\pi n} \cos\left(\frac{\pi n}{3}\right) = \frac{1}{2}\left[e^{j\frac{4\pi}{3}n} + e^{j\frac{2\pi}{3}n}\right]$$

Therefore:

$$X_4(z) = \frac{1}{2}\left[\frac{1}{1-e^{j\frac{4\pi}{3}}z^{-1}} + \frac{1}{1-e^{j\frac{2\pi}{3}}z^{-1}}\right], \quad \text{ROC: } |z| > 1$$

(e) $x(n) = (-1)^n u(n)$

$$X(z) = \sum_{n=0}^{\infty} (-1)^n z^{-n} = \frac{1}{1+z^{-1}}, \quad \text{ROC: } |z| > 1$$

(f) $x(n) = \{\underset{\uparrow}{1}, 0, -1, 0, 1, -1, \dots\}$

$$X(z) = 1 - z^{-2} + z^{-4} - z^{-5}, z \neq 0$$

1.5.3 Exercise 3.7

Compute the convolution of the following signals by means of the z-transform.

$$x_1(n) = \begin{cases} (\frac{1}{3})^n, & n \geq 0 \\ (\frac{1}{2})^{-n}, & n < 0 \end{cases}$$

$$x_2(n) = (\frac{1}{2})^n u(n)$$

Solution

$$
\begin{aligned}
X_1(z) &= \frac{z}{z-1/3} - \frac{z}{z-2}, \quad \frac{1}{3} < |z| < 2 \\
X_2(z) &= \frac{z}{z-1/2}, \quad |z| > \frac{1}{2} \\
\frac{Y(z)}{z} &= \frac{z}{(z-1/3)(z-1/2)} - \frac{z}{(z-2)(z-1/2)} \\
&= \left(\frac{A}{z-1/3} + \frac{B}{z-1/2}\right) - \left(\frac{C}{z-2} + \frac{D}{z-1/2}\right) \\
A &= \left.\frac{z}{z-1/2}\right|_{z=1/3} = \frac{1/3}{-1/6} = -2 \\
B &= \left.\frac{z}{z-1/3}\right|_{z=1/2} = \frac{1/2}{1/6} = 3 \\
C &= \left.\frac{z}{z-1/2}\right|_{z=2} = \frac{2}{3/2} = \frac{4}{3} \\
D &= \left.\frac{z}{z-2}\right|_{z=1/2} = \frac{1/2}{-3/2} = -\frac{1}{3} \\
\frac{Y(z)}{z} &= \frac{-2}{z-1/3} + \frac{3-(-1/3)}{z-1/2} - \frac{4/3}{z-2} = \frac{-2}{z-1/3} + \frac{10/3}{z-1/2} - \frac{4/3}{z-2} \\
Y(z) &= -2\left(\frac{z}{z-1/3}\right) + \frac{10}{3}\left(\frac{z}{z-1/2}\right) - \frac{4}{3}\left(\frac{z}{z-2}\right) \\
y(n) &= \left[-2\left(\frac{1}{3}\right)^n + \frac{10}{3}\left(\frac{1}{2}\right)^n\right]u(n) + \frac{4}{3}(2)^n u(-n-1)
\end{aligned}
$$

1.5.4 Exercise 3.8

Use the convolution property to:

(a) Express the z-transform of

$$y(n) = \sum_{k=-\infty}^{n} x(k)$$

in terms of $X(z)$.

(b) Determine the z-transform of $x(n) = (n+1)u(n)$. (Hint: Show first that $x(n) = u(n) * u(n)$.)

Solution

(a)

$$y(n) = \sum_{k=-\infty}^{n} x(k)$$

We can rewrite the accumulation as a convolution with the unit step function $u(n)$:

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)u(n-k) = x(n) * u(n)$$

Applying the convolution property of the z-transform ($Y(z) = X(z)U(z)$) and knowing $U(z) = \frac{z}{z-1} = \frac{1}{1-z^{-1}}$ for $|z| > 1$:

$$Y(z) = X(z)\left(\frac{z}{z-1}\right) = \frac{X(z)}{1-z^{-1}}$$

(b)

$$\text{First, prove the hint: } u(n) * u(n) = \sum_{k=-\infty}^{\infty} u(k)u(n-k)$$

Since $u(k) = 1$ for $k \geq 0$ and $u(n-k) = 1$ for $k \leq n$, the non-zero terms exist only for $0 \leq k \leq n$ (which means $n \geq 0$):

$$\begin{aligned} u(n) * u(n) &= \left(\sum_{k=0}^{n} 1\right) u(n) \\ &= (n - 0 + 1)u(n) = (n+1)u(n) = x(n) \end{aligned}$$

Now, apply the convolution property to find $X(z)$:

$$\begin{aligned} X(z) &= \mathcal{Z}\{u(n) * u(n)\} = U(z) \cdot U(z) \\ X(z) &= \left(\frac{z}{z-1}\right)\left(\frac{z}{z-1}\right) = \frac{z^2}{(z-1)^2}, \quad |z| > 1 \end{aligned}$$

1.5.5 Exercise 3.13

Let $x(n)$ be a sequence with z-transform $X(z)$. Determine, in terms of $X(z)$, the z-transforms of the following signals.

(a) $x_1(n) = \begin{cases} x\left(\frac{n}{2}\right), & \text{if } n \text{ even} \\ 0, & \text{if } n \text{ odd} \end{cases}$

(b) $x_2(n) = x(2n)$

Solution

(a)

Given:

$$x_1(n) = \begin{cases} x\left(\frac{n}{2}\right), & \text{if } n \text{ even} \\ 0, & \text{if } n \text{ odd} \end{cases}$$

The z-transform definition is:

$$X_1(z) = \sum_{n=-\infty}^{\infty} x_1(n) z^{-n}$$

Since $x_1(n) = 0$ for odd n, we sum only over even indices. Let $n = 2k$:

$$\begin{aligned} X_1(z) &= \sum_{k=-\infty}^{\infty} x_1(2k) z^{-2k} \\ &= \sum_{k=-\infty}^{\infty} x\left(\frac{2k}{2}\right) z^{-2k} \\ &= \sum_{k=-\infty}^{\infty} x(k)(z^2)^{-k} \\ X_1(z) &= X(z^2) \end{aligned}$$

(b)

Given:

$$x_2(n) = x(2n)$$

Define an intermediate sequence $v(n)$ that retains only the even-indexed samples of $x(n)$ using the term $\frac{1+(-1)^n}{2}$:

$$v(n) = x(n)\left[\frac{1+(-1)^n}{2}\right]$$

Taking the z-transform of $v(n)$:

$$\begin{aligned} V(z) &= \sum_{n=-\infty}^{\infty} x(n)\left[\frac{1+(-1)^n}{2}\right] z^{-n} \\ &= \frac{1}{2}\sum_{n=-\infty}^{\infty} x(n) z^{-n} + \frac{1}{2}\sum_{n=-\infty}^{\infty} x(n)(-1)^n z^{-n} \\ V(z) &= \frac{1}{2}\left[X(z) + X(-z)\right] \end{aligned}$$

Alternatively, by the definition of $V(z)$ and knowing $v(n) = 0$ for odd n, we sum over $n = 2k$:

$$\begin{aligned} V(z) &= \sum_{k=-\infty}^{\infty} v(2k) z^{-2k} \\ &= \sum_{k=-\infty}^{\infty} x(2k) z^{-2k} \\ &= \sum_{k=-\infty}^{\infty} x_2(k) (z^2)^{-k} \\ V(z) &= X_2(z^2) \end{aligned}$$

Equating the two expressions for $V(z)$:

$$X_2(z^2) = \frac{1}{2}\left[X(z) + X(-z)\right]$$

Substituting z with $z^{1/2}$ yields the final transform:

$$X_2(z) = \frac{1}{2}\left[X(z^{1/2}) + X(-z^{1/2})\right]$$

1.5.6 Exercise 3.14

Determine the causal signal $x(n)$ if its z-transform $X(z)$ is given by:

(a) $X(z) = \frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}}$

(b) $X(z) = \frac{1}{1-z^{-1}+\frac{1}{2}z^{-2}}$

(c) $X(z) = \frac{z^{-6}+z^{-7}}{1-z^{-1}}$

(d) $X(z) = \frac{1+2z^{-2}}{1+z^{-2}}$

(e) $X(z) = \frac{1}{4}\frac{1+6z^{-1}+z^{-2}}{(1-2z^{-1}+2z^{-2})(1-0.5z^{-1})}$

(f) $X(z) = \frac{2-1.5z^{-1}}{1-1.5z^{-1}+0.5z^{-2}}$

(g) $X(z) = \frac{1+2z^{-1}+z^{-2}}{1+4z^{-1}+4z^{-2}}$

(h) $X(z)$ is specified by a pole–zero pattern in Fig. P3.14. The constant $G = \frac{1}{4}$.

(i) $X(z) = \frac{1-\frac{1}{2}z^{-1}}{1+\frac{1}{2}z^{-1}}$

(j) $X(z) = \frac{1-az^{-1}}{z^{-1}-a}$

Solution

(a) $X(z) = \frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}}$

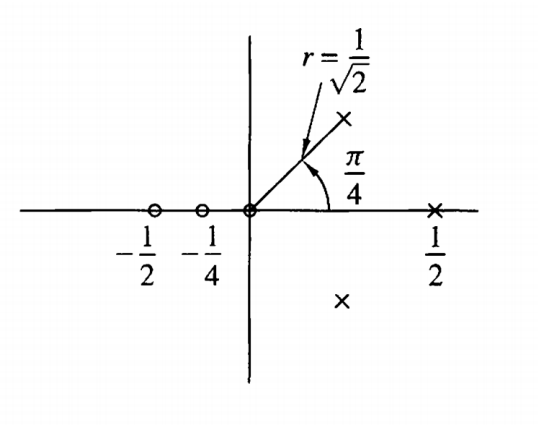

Figure 7: *Fig P3.14*

First, we use partial fraction expansion on $X(z)$:

$$X(z) = \frac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}} = \frac{1+3z^{-1}}{(1+z^{-1})(1+2z^{-1})}$$
$$X(z) = \frac{A}{1+z^{-1}} + \frac{B}{1+2z^{-1}}$$

We solve for the coefficients A and B by evaluating at the poles:

$$A = \left.\frac{1+3z^{-1}}{1+2z^{-1}}\right|_{z^{-1}=-1} = \frac{1-3}{1-2} = 2$$
$$B = \left.\frac{1+3z^{-1}}{1+z^{-1}}\right|_{z^{-1}=-0.5} = \frac{1-1.5}{1-0.5} = -1$$

Applying the standard inverse transform $\mathcal{Z}^{-1}\left\{\frac{1}{1-az^{-1}}\right\} = a^n u(n)$, we obtain:

$$x(n) = \left[2(-1)^n - (-2)^n\right]u(n)$$

(b) $X(z) = \frac{1}{1-z^{-1}+\frac{1}{2}z^{-2}}$

We rewrite $X(z)$ to match the standard form for exponentially damped sinusoids:

$$X(z) = \frac{1}{1-z^{-1}+0.5z^{-2}}$$
$$= \frac{1-0.5z^{-1}}{1-z^{-1}+0.5z^{-2}} + \frac{0.5z^{-1}}{1-z^{-1}+0.5z^{-2}}$$

Comparing the denominator to $1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}$, we find the radius r and frequency

ω_0:

$$r^2 = 0.5 \implies r = \frac{1}{\sqrt{2}}$$
$$2r\cos(\omega_0) = 1 \implies \cos(\omega_0) = \frac{1}{\sqrt{2}} \implies \omega_0 = \frac{\pi}{4}$$

Taking the inverse transform of both the cosine and sine components yields:

$$x(n) = \left(\frac{1}{\sqrt{2}}\right)^n \left[\cos\left(\frac{\pi}{4}n\right) + \sin\left(\frac{\pi}{4}n\right)\right] u(n)$$

(c) $X(z) = \frac{z^{-6}+z^{-7}}{1-z^{-1}}$

We separate the terms and apply the time-shifting property $z^{-k}X(z) \leftrightarrow x(n-k)$:

$$X(z) = z^{-6}\left(\frac{1}{1-z^{-1}}\right) + z^{-7}\left(\frac{1}{1-z^{-1}}\right)$$

Since the inverse transform of $\frac{1}{1-z^{-1}}$ is $u(n)$, shifting by 6 and 7 samples gives:

$$x(n) = u(n-6) + u(n-7)$$

(d) $X(z) = \frac{1+2z^{-2}}{1+z^{-2}}$

First, decompose the numerator to simplify the expression:

$$X(z) = \frac{1+z^{-2}+z^{-2}}{1+z^{-2}} = 1 + z^{-2}\left(\frac{1}{1+z^{-2}}\right)$$

We recognize the standard cosine transform where $\omega_0 = \pi/2$:

$$\mathcal{Z}^{-1}\left\{\frac{1}{1+z^{-2}}\right\} = \cos\left(\frac{\pi}{2}n\right) u(n)$$

Applying the inverse transform and the time-shifting property for the z^{-2} multiplier, we get:

$$x(n) = \delta(n) + \cos\left(\frac{\pi}{2}(n-2)\right) u(n-2)$$

(e) $X(z) = \frac{1}{4}\frac{1+6z^{-1}+z^{-2}}{(1-2z^{-1}+2z^{-2})(1-0.5z^{-1})}$

We use partial fraction expansion for the real pole and the complex conjugate poles:

$$\begin{aligned} X(z) &= \frac{0.25 + 1.5z^{-1} + 0.25z^{-2}}{(1-2z^{-1}+2z^{-2})(1-0.5z^{-1})} \\ &= \frac{-0.6 + 2.9z^{-1}}{1-2z^{-1}+2z^{-2}} + \frac{0.85}{1-0.5z^{-1}} \end{aligned}$$

By splitting the complex pole term into standard cosine and sine transform pairs ($r = \sqrt{2}, \omega_0 = \pi/4$), we take the inverse transform:

$$x(n) = \left[0.85(0.5)^n + (\sqrt{2})^n \left(-0.6\cos\left(\frac{\pi}{4}n\right) + 2.3\sin\left(\frac{\pi}{4}n\right)\right)\right] u(n)$$

(f) $X(z) = \frac{2-1.5z^{-1}}{1-1.5z^{-1}+0.5z^{-2}}$

First, factor the denominator and expand into partial fractions:

$$\begin{aligned} X(z) &= \frac{2-1.5z^{-1}}{1-1.5z^{-1}+0.5z^{-2}} \\ &= \frac{2-1.5z^{-1}}{(1-z^{-1})(1-0.5z^{-1})} = \frac{1}{1-z^{-1}} + \frac{1}{1-0.5z^{-1}} \end{aligned}$$

Applying the standard inverse transform $\mathcal{Z}^{-1}\left\{\frac{1}{1-az^{-1}}\right\} = a^n u(n)$ yields:

$$x(n) = \left[1 + (0.5)^n\right] u(n)$$

(g) $X(z) = \frac{1+2z^{-1}+z^{-2}}{1+4z^{-1}+4z^{-2}}$

We use the algebraic decomposition to separate the terms:

$$\begin{aligned} X(z) &= \frac{1+2z^{-1}+z^{-2}}{1+4z^{-1}+4z^{-2}} \\ &= 1 - \left(\frac{2z^{-1}+3z^{-2}}{(1+2z^{-1})^2}\right) \\ &= 1 - \frac{2z^{-1}}{1+2z^{-1}} + \frac{z^{-2}}{(1+2z^{-1})^2} \end{aligned}$$

Next, we apply the inverse Z-transforms. Note the correct inverse for the last term introduces a $-\frac{1}{2}$ multiplier:

$$\begin{aligned} \mathcal{Z}^{-1}\{1\} &= \delta(n) \\ \mathcal{Z}^{-1}\left\{\frac{2z^{-1}}{1+2z^{-1}}\right\} &= 2(-2)^{n-1}u(n-1) \\ \mathcal{Z}^{-1}\left\{\frac{z^{-2}}{(1+2z^{-1})^2}\right\} &= -\frac{1}{2}(n-1)(-2)^{n-1}u(n-1) \end{aligned}$$

Combining these and factoring out $(-2)^{n-1}u(n-1)$ gives the correct causal signal:

$$x(n) = \delta(n) - 2(-2)^{n-1}u(n-1) - \frac{1}{2}(n-1)(-2)^{n-1}u(n-1)$$
$$x(n) = \delta(n) - \left[2 + \frac{1}{2}(n-1)\right](-2)^{n-1}u(n-1)$$
$$x(n) = \delta(n) - 0.5(n+3)(-2)^{n-1}u(n-1)$$

(h) $X(z)$ is specified by a pole–zero pattern in Fig. P3.14. The constant $G = \frac{1}{4}$.

First, we extract the poles and zeros from the given plot. We have zeros at $z = -1/2$, $z = -1/4$, and poles at $z = 1/2$, $z = \frac{1}{\sqrt{2}}e^{\pm j\pi/4}$. Given the constant $G = 1/4$, we can write the transfer function as:

$$\begin{aligned} X(z) &= \frac{1}{4}\frac{(z+1/2)(z+1/4)}{(z-1/2)(z-\frac{1}{\sqrt{2}}e^{j\pi/4})(z-\frac{1}{\sqrt{2}}e^{-j\pi/4})} \\ &= \frac{1}{4}\frac{z^2+\frac{3}{4}z+\frac{1}{8}}{(z-1/2)(z^2-z+1/2)} \end{aligned}$$

To convert this to negative powers of z, we multiply the numerator and denominator by z^{-3}. We purposefully factor out a z^{-1} in the numerator to align the remaining terms with standard $\mathcal{Z}$-transform pairs:

$$X(z) = \frac{1}{4}\frac{(1+\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2})z^{-1}}{(1-\frac{1}{2}z^{-1})(1-z^{-1}+\frac{1}{2}z^{-2})}$$

Next, we perform a partial fraction expansion on the main fraction (ignoring the z^{-1} for a moment). We structure the complex pole terms to perfectly match the standard exponentially damped cosine and sine forms (where $r = 1/\sqrt{2}$ and $\omega_0 = \pi/4$):

$$\frac{1}{4}\frac{1+\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}{(1-\frac{1}{2}z^{-1})(1-z^{-1}+\frac{1}{2}z^{-2})} = \frac{A(1-\frac{1}{2}z^{-1})}{1-z^{-1}+\frac{1}{2}z^{-2}} + \frac{B(\frac{1}{2}z^{-1})}{1-z^{-1}+\frac{1}{2}z^{-2}} + \frac{C}{1-\frac{1}{2}z^{-1}}$$

By finding a common denominator and equating the polynomial coefficients, we solve for A, B, and C:

$$A = -\frac{1}{2}, \quad B = \frac{7}{8}, \quad C = \frac{3}{4}$$

Substituting these coefficients back into our equation for $X(z)$ gives:

$$X(z) = z^{-1}\left[\frac{-\frac{1}{2}(1-\frac{1}{2}z^{-1})}{1-z^{-1}+\frac{1}{2}z^{-2}} + \frac{\frac{7}{8}(\frac{1}{2}z^{-1})}{1-z^{-1}+\frac{1}{2}z^{-2}} + \frac{\frac{3}{4}}{1-\frac{1}{2}z^{-1}}\right]$$

The terms in the brackets correspond to $-\frac{1}{2}r^n\cos(\omega_0 n)$, $\frac{7}{8}r^n\sin(\omega_0 n)$, and $\frac{3}{4}(\frac{1}{2})^n$.

Finally, the z^{-1} multiplier applies a time shift of $(n-1)$ to the entire signal. Noting that the radius $r^{n-1} = \left(\frac{1}{\sqrt{2}}\right)^{n-1} = \left(\frac{1}{2}\right)^{\frac{n-1}{2}}$, we apply the inverse $\mathcal{Z}$-transform to get the causal signal:

$$x(n) = \left[-\frac{1}{2}\left(\frac{1}{2}\right)^{\frac{n-1}{2}} \cos\left(\frac{\pi}{4}(n-1)\right) + \frac{7}{8}\left(\frac{1}{2}\right)^{\frac{n-1}{2}} \sin\left(\frac{\pi}{4}(n-1)\right) + \frac{3}{4}\left(\frac{1}{2}\right)^{n-1}\right] u(n-1)$$

(i) $X(z) = \frac{1-\frac{1}{2}z^{-1}}{1+\frac{1}{2}z^{-1}}$

We split the numerator into two distinct terms to apply standard transforms:

$$\begin{aligned} X(z) &= \frac{1-\frac{1}{4}z^{-1}}{1+\frac{1}{2}z^{-1}} \\ &= \frac{1}{1+\frac{1}{2}z^{-1}} - \frac{1}{4}\left(\frac{z^{-1}}{1+\frac{1}{2}z^{-1}}\right) \end{aligned}$$

Applying the standard inverse transform $\mathcal{Z}^{-1}\left\{\frac{1}{1-az^{-1}}\right\} = a^n u(n)$ alongside the time-shifting property for the second term, we retain the minus sign:

$$x(n) = \left(-\frac{1}{2}\right)^n u(n) - \frac{1}{4}\left(-\frac{1}{2}\right)^{n-1} u(n-1)$$

Alternative Form: If we had instead extracted a constant term using polynomial division (similar to how we approached problem g), we would get an equivalent, completely valid form:

$$\begin{aligned} X(z) &= -0.5 + \frac{1.5}{1+\frac{1}{2}z^{-1}} \\ x(n) &= -0.5\delta(n) + 1.5\left(-\frac{1}{2}\right)^n u(n) \end{aligned}$$

(j) $X(z) = \frac{1-az^{-1}}{z^{-1}-a}$

First, we factor out $-a$ from the denominator to get it into the standard $1-\alpha z^{-1}$ format:

$$X(z) = \frac{1-az^{-1}}{-a+z^{-1}} = \frac{1-az^{-1}}{-a(1-a^{-1}z^{-1})}$$

Now, we separate the numerator terms to create two standard transform expressions:

$$\begin{aligned} X(z) &= -\frac{1}{a}\left(\frac{1}{1-a^{-1}z^{-1}}\right) + \frac{az^{-1}}{a(1-a^{-1}z^{-1})} \\ &= -\frac{1}{a}\left(\frac{1}{1-a^{-1}z^{-1}}\right) + z^{-1}\left(\frac{1}{1-a^{-1}z^{-1}}\right) \end{aligned}$$

Taking the inverse $\mathcal{Z}$-transform of each term (using the time-shifting property on the second

term) yields the causal signal:

$$x(n) = -\frac{1}{a}(a^{-1})^n u(n) + (a^{-1})^{n-1} u(n-1)$$

Note: This can also be validly simplified to $x(n) = -(a)^{-(n+1)}u(n) + (a)^{-(n-1)}u(n-1)$.

1.5.7 Exercise 3.19

By first differentiating $X(z)$ and then using appropriate properties of the z-transform, determine $x(n)$ for the following transforms.

(a) $X(z) = \log(1-2z), \quad |z| < \frac{1}{2}$

(b) $X(z) = \log(1-z^{-1}), \quad |z| > \frac{1}{2}$

Solution

(a) $X(z) = \log(1-2z), \quad |z| < \frac{1}{2}$

We use the differentiation property of the Z-transform, which states that $Z\{nx(n)\} = -z\frac{dX(z)}{dz}$. Let's define $Y(z) = -z\frac{dX(z)}{dz}$:

$$\begin{aligned}
X(z) &= \log(1-2z), \quad |z| < \frac{1}{2} \\
Y(z) &= -z\frac{dX(z)}{dz} \\
&= -z\left(\frac{-2}{1-2z}\right) = \frac{2z}{1-2z} \\
&= \frac{-1}{1-\frac{1}{2}z^{-1}}, \quad |z| < \frac{1}{2}
\end{aligned}$$

Since the Region of Convergence (ROC) is $|z| < \frac{1}{2}$, it corresponds to a left-sided sequence. Taking the inverse Z-transform yields:

$$y(n) = \left(\frac{1}{2}\right)^n, \quad n < 0$$

We can express this using the unit step function as $y(n) = \left(\frac{1}{2}\right)^n u(-n-1)$.

Because $Y(z)$ is the transform of $nx(n)$, it follows that $x(n) = \frac{1}{n}y(n)$:

$$\begin{aligned}
x(n) &= \frac{1}{n}y(n) \\
&= \frac{1}{n}\left(\frac{1}{2}\right)^n u(-n-1)
\end{aligned}$$

(b) $X(z) = \log(1-z^{-1}), \quad |z| > \frac{1}{2}$

First, we use the differentiation property of the Z-transform, which states that $Y(z) = \mathcal{Z}\{nx(n)\} = -z\frac{dX(z)}{dz}$.

$$\begin{aligned} X(z) &= \log(1 - z^{-1}), \quad |z| > \frac{1}{2} \\ \frac{dX(z)}{dz} &= \frac{1}{1 - z^{-1}} \cdot \frac{d}{dz}(1 - z^{-1}) \\ &= \frac{1}{1 - z^{-1}} \cdot (z^{-2}) = \frac{z^{-2}}{1 - z^{-1}} \end{aligned}$$

Now, multiply by $-z$ to find $Y(z)$:

$$\begin{aligned} Y(z) &= -z\left(\frac{z^{-2}}{1 - z^{-1}}\right) \\ &= \frac{-z^{-1}}{1 - z^{-1}} \end{aligned}$$

The given ROC indicates a right-sided sequence. We use the standard Z-transform pair $\mathcal{Z}\{u(n)\} = \frac{1}{1-z^{-1}}$ and apply the time-shifting property due to the z^{-1} in the numerator ($\mathcal{Z}\{x(n-1)\} = z^{-1}X(z)$):

$$\begin{aligned} y(n) &= -1 \cdot u(n-1) \\ &= -u(n-1) \end{aligned}$$

Since $Y(z)$ is the transform of $nx(n)$, we simply divide by n to find our final sequence $x(n)$:

$$\begin{aligned} x(n) &= \frac{1}{n}y(n) \\ &= -\frac{1}{n}u(n-1) \end{aligned}$$

1.5.8 Exercise 3.20

(a) Draw the pole–zero pattern for the signal

$$x_1(n) = (r^n \sin \omega_0 n)u(n), \quad 0 < r < 1$$

(b) Compute the z-transform $X_2(z)$, which corresponds to the pole–zero pattern in part (a).

(c) Compare $X_1(z)$ with $X_2(z)$. Are they identical? If not, indicate a method to derive $X_1(z)$ from the pole–zero pattern.

Solution

(a) Draw the pole–zero pattern for the signal $x_1(n) = (r^n \sin \omega_0 n)u(n), \quad 0 < r < 1$

First, we find the Z-transform of the given signal $x_1(n)$. Using standard Z-transform tables for a exponentially damped sine wave:

$$x_1(n) = r^n \sin(\omega_0 n)u(n), \quad 0 < r < 1$$
$$X_1(z) = \frac{r \sin(\omega_0) z^{-1}}{1 - 2r\cos(\omega_0)z^{-1} + r^2 z^{-2}}$$

This is the Scilab code to draw pole-zero pattern:

```
// 1. Define example parameters (you can change these!)
r = 0.8;           // Radius constraint: 0 < r < 1
w0 = %pi / 4;      // Frequency (e.g., pi/4)

// 2. Define the polynomial variable 'z'
z = poly(0, 'z');

// 3. Define the Numerator and Denominator polynomials
num = r * sin(w0) * z;
den = z^2 - 2 * r * cos(w0) * z + r^2;

// 4. Create the discrete-time linear transfer function
// 'd' stands for discrete time
H = syslin('d', num, den);

// 5. Plot the pole-zero map
clf();             // Clear the current figure window
plzr(H);           // The standard Scilab function for Pole-Zero plots

// 6. Formatting the plot
title( Pole-Zero Pattern for X_1(z) );
```

This is the result:

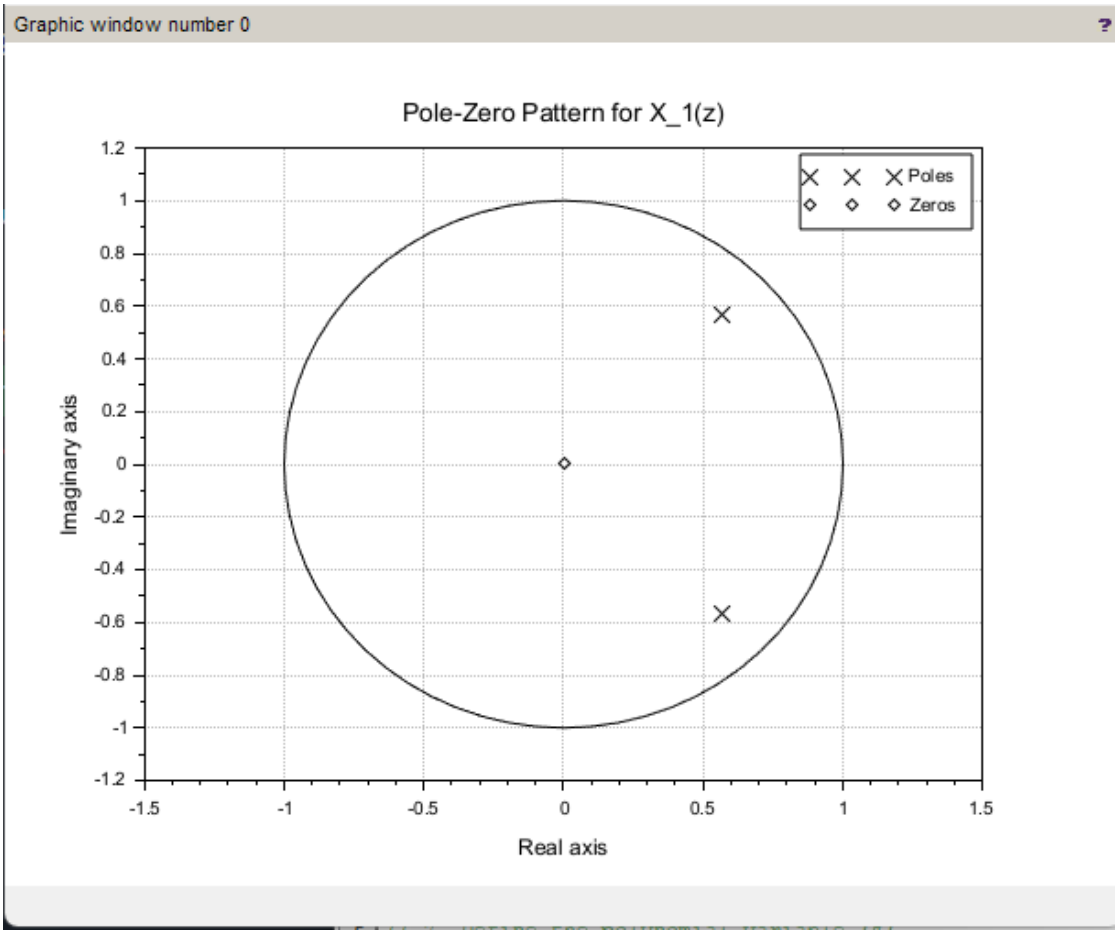

Figure 8: *Pole–zero pattern for the signal $x_1(n)$*

(b) Compute the z-transform $X_2(z)$, which corresponds to the pole–zero pattern in part (a).

Now, we compute a new Z-transform $X_2(z)$ based strictly on the pole-zero pattern from part (a). A system with poles at $re^{\pm j\omega_0}$ and a zero at $z = 0$ has the general form:

$$\begin{aligned} X_2(z) &= \frac{z}{(1 - re^{j\omega_0}z^{-1})(1 - re^{-j\omega_0}z^{-1})} \\ &= \frac{z}{1 - r(e^{j\omega_0} + e^{-j\omega_0})z^{-1} + r^2 e^{j\omega_0}e^{-j\omega_0}z^{-2}} \end{aligned}$$

Using Euler's formula ($e^{j\omega_0} + e^{-j\omega_0} = 2\cos\omega_0$), this simplifies to:

$$X_2(z) = \frac{z}{1 - 2r\cos(\omega_0)z^{-1} + r^2 z^{-2}}$$

(c) Compare $X_1(z)$ with $X_2(z)$. Are they identical? If not, indicate a method to derive $X_1(z)$ from the pole–zero pattern.

Comparing $X_1(z)$ from part (a) and $X_2(z)$ from part (b):

- $X_1(z)$ and $X_2(z)$ share the same denominator (same poles).
- They differ in the numerator. The pole-zero pattern only gives us the location of the roots, but it does not account for any constant scaling factors.

Conclusion: $X_1(z)$ and $X_2(z)$ differ by a constant factor. This scaling constant can be determined by evaluating and matching the values of $X_1(z)$ and the unscaled transfer function at a specific point, such as $z = 1$.

2 Reference

The full source code and this report is available in the GitHub repository: https://github.com/l1ttled1no/DigitalSignalProcessing-Lab